

HAIL, Saudi Arabia

WMO: 403940

Lat: 27.438N

Lon: 41.686E

Elev: 1015

StdP: 89.71

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.8	2.7	-12.1	1.5	13.0	-9.9	1.8	13.1	12.0	18.3	10.5	17.0	1.5	180	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.5	42.1	18.7	41.2	18.3	40.2	18.0	19.7	38.8	19.2	38.3	18.7	37.9	4.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	12.2	20.4	14.0	11.3	20.0	12.8	10.4	20.2	60.8	38.6	58.5	38.5	56.9	37.5	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.4	9.3	8.0	-3.4	43.5	2.3	0.8	-5.1	44.0	-6.5	44.5	-7.8	45.0	-9.5	45.5	
			DB												
			WB	-5.2	22.0	2.3	1.8	-6.8	23.3	-8.2	24.3	-9.4	25.3	-11.1	26.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	11.4	13.6	17.3	22.6	27.6	31.1	32.1	32.8	30.2	24.9	17.4	12.8
	DBStd	8.26	3.57	3.67	3.71	3.23	2.69	1.90	2.09	2.00	2.21	2.64	3.49	3.30
	HDD10.0	48	26	8	1	0	0	0	0	0	0	0	1	12
	HDD18.3	654	217	138	63	5	0	0	0	0	0	1	58	172
	CDD10.0	4744	68	108	228	377	547	632	686	708	606	463	222	98
	CDD18.3	2309	1	4	33	132	289	382	428	449	356	204	29	1
	CDH23.3	36656	21	90	473	1566	4092	6394	7478	7857	5646	2639	363	36
	CDH26.7	23438	1	13	151	687	2375	4282	5203	5554	3687	1389	94	1
Wind	WSAvg	3.3	3.2	3.5	3.7	3.9	3.8	3.5	3.4	3.0	2.9	3.1	3.0	3.0
Precipitation	PrecAvg	85	12	6	22	11	7	0	0	0	0	7	12	7
	PrecMax	245	41	32	83	47	40	2	4	1	1	30	46	55
	PrecMin	11	0	0	2	0	0	0	0	0	0	1	0	0
	PrecStd	45	11	11	26	12	10	0	1	0	0	9	15	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.9	28.6	33.1	36.1	39.9	41.9	43.0	43.1	41.2	37.0	31.0	26.1
		MCWB	13.0	13.9	14.9	16.0	17.5	18.1	18.8	19.4	18.3	17.1	15.4	12.8
	2%	DB	23.2	26.0	30.3	34.2	38.2	40.9	42.0	42.1	40.2	35.8	29.2	24.2
		MCWB	11.9	13.2	14.2	16.1	17.1	17.8	18.6	19.0	18.0	16.5	14.3	13.0
	5%	DB	21.2	24.0	28.1	32.8	36.9	40.0	41.0	41.3	39.2	34.8	27.5	22.2
		MCWB	11.2	11.9	13.5	15.5	16.8	17.6	18.2	18.6	17.7	16.0	14.0	12.1
	10%	DB	19.0	21.9	25.9	30.8	35.7	38.9	40.0	40.8	38.1	33.2	25.2	20.2
		MCWB	10.2	11.0	12.7	14.9	16.5	17.1	17.8	18.4	17.3	15.5	13.4	11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	15.8	17.1	18.9	19.8	20.1	20.1	21.0	19.6	18.9	17.7	16.3
		MCDB	22.3	24.1	26.5	27.8	33.5	38.3	40.4	40.1	38.8	30.0	24.1	22.2
	2%	WB	13.6	14.1	15.5	17.6	18.7	18.7	19.2	19.9	18.7	17.6	16.6	14.5
		MCDB	20.5	22.8	26.4	28.8	33.1	38.9	40.0	40.3	38.3	30.4	23.6	21.2
	5%	WB	12.2	13.0	14.4	16.7	17.8	18.1	18.7	19.2	18.2	16.8	15.7	13.3
		MCDB	18.9	21.9	25.4	28.1	33.1	38.5	39.5	39.7	38.0	30.8	22.7	19.9
	10%	WB	10.9	11.7	13.3	15.9	17.1	17.5	18.1	18.8	17.6	16.2	14.7	12.0
		MCDB	17.6	20.5	24.3	27.7	32.7	37.6	38.9	39.3	37.1	30.8	22.0	18.4
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.0	14.2	14.3	14.4	15.4	15.1	15.5	15.9	15.7	13.4	13.4
		MCDBR	15.8	16.3	16.7	16.2	15.8	16.3	15.9	15.9	16.3	16.8	16.3	15.9
		MCWBR	8.2	8.4	7.8	6.5	6.3	6.4	6.3	6.3	6.5	7.1	7.4	8.3
	5% WB	MCDBR	13.6	14.4	14.9	14.2	14.7	15.3	15.1	15.1	15.5	15.2	12.7	13.5
		MCWBR	7.8	7.9	7.5	6.3	6.1	6.5	6.3	6.3	6.4	6.5	6.3	7.6
Clear-Sky Solar Irradiance	taub	0.341	0.383	0.447	0.521	0.543	0.492	0.469	0.510	0.476	0.468	0.416	0.365	
	taud	2.254	2.105	1.909	1.763	1.727	1.844	1.937	1.827	1.884	1.894	2.026	2.170	
	Ebn at Noon	896	882	843	792	772	809	826	790	805	780	798	845	
	Edh at Noon	117	148	191	227	236	209	190	211	193	180	147	123	
All-Sky Solar Radiation	RadAvg	4.21	5.12	6.19	6.87	7.45	8.22	8.00	7.42	6.57	5.44	4.27	3.86	
	RadStd	0.24	0.22	0.21	0.21	0.31	0.20	0.22	0.20	0.20	0.18	0.27	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.49	N/A	N/A	+0.24	N/A	N/A	N/A	-52	+215	+165
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page